CMPE 492 “TEDU GuidAR” Senior Design Project Backlog

Task Name	User Story	Sprint Ready	Priority	Status	Effort	Assigned to Sprint
SPRINT 6	No	No	Not Started	In Progress	86	Yes
Test Plan Design & Documentation	As a tester, I want a formal test plan so that all system components are evaluated systematically	Yes	High	In Progress	10	Yes
Unit Testing for Core Modules	As a developer, I want to test individual modules (SLAM, path planning, AR rendering) so that bugs are isolated early	Yes	High	In Progress	12	Yes
Integration Testing (Module Interactions)	As a developer, I want to verify interactions between modules so that the full system behaves correctly	Yes	High	In Progress	14	Yes
Stress Testing in Complex Indoor Environments	As a tester, I want to evaluate performance in crowded or complex spaces so that system limits are identified	Yes	High	Not Started	12	Yes
Tracking Loss & Recovery Testing	As a user, I want the system to recover quickly from tracking loss so that navigation is uninterrupted	Yes	High	Not Started	10	Yes
Multi-Floor Navigation Testing	As a user, I want accurate guidance across floors so that vertical navigation works reliably	No	Medium	Not Started	8	Yes
Performance Testing (FPS, Latency, Resource Usage)	As a developer, I want to measure system performance under different conditions so that optimization targets are clear	Yes	High	In Progress	12	Yes
Bug Tracking & Issue Prioritization	As a team, we want to log and prioritize bugs so that critical issues are addressed first	Yes	High	In Progress	8	Yes

SPRINT 5	No	No	Not Started	In Progress	80	Yes
Initial Real-World Environment Testing	Test the system in a real building.	Yes	High	In Progress	16	Yes
Navigation Accuracy Benchmarking	Measure localization and routing accuracy.	No	Medium	Not Started	8	Yes
Basic UX Refinement	Make clearer navigation cues.	Yes	High	In Progress	10	Yes
Performance Profiling & Bottleneck Identification	Identify FPS drops and resource bottlenecks.	Yes	Medium	In Progress	8	Yes
Tracking Stability Improvements	Ensure more stable tracking.	Yes	Medium	In Progress	8	Yes
Test Scenario Development	Define indoor test scenarios.	No	High	Not Started	12	Yes
Data Logging & Debug Tools	Log SLAM and navigation.	Yes	High	In Progress	10	Yes
Refinement Backlog Creation	Document issues and improvements.	Yes	High	In Progress	8	Yes
SPRINT 4	No	Yes	In Progress	In Progress	80	Yes
Advanced Multi-Floor Routing Logic	Improve routing decisions across complex floor transitions.	Yes	High	In Progress	12	Yes
Landmark-Based Navigation Support	Use visual landmarks to enhance navigation clarity.	Yes	Medium	In Progress	8	Yes
Re-localization After Tracking Loss	Recover user position after SLAM failure.	Yes	High	In Progress	12	Yes
Dynamic Obstacle Awareness	Adjust navigation cues based on moving obstacles.	No	Medium	In Progress	10	Yes
User Feedback Collection Module	Allow users to report navigation issues.	Yes	Medium	In Progress	6	Yes
Map Scaling for Large Buildings	Support larger and more complex indoor spaces.	Yes	High	In Progress	14	Yes

Indoor POI (Point of Interest) Management	Add, update, and categorize indoor locations.	Yes	High	In Progress	8	Yes
Continuous Localization Accuracy Monitoring	Track navigation accuracy over time.	Yes	High	Not Started	10	Yes
SPRINT 3	No	Yes	High	Complete	114	Yes
Cross-Platform AR Framework Integration	Enable consistent AR behavior across Meta Quest 3 and mobile devices.	Yes	High	In Progress	16	Yes
Persistent Spatial Anchors	Allow users to revisit the same indoor map accurately.	Yes	High	In Progress	14	Yes
Improved Occlusion Handling	Ensure navigation cues respect real-world obstacles.	Yes	Medium	Not Started	10	Yes
Map Persistence & Storage	Save and reload indoor maps.	Yes	High	In Progress	12	Yes
User Calibration Flow	Guide users through initial calibration process.	Yes	Medium	In Progress	8	Yes
Gesture-Based Interaction	Enable gesture controls for navigation actions.	Yes	Medium	In Progress	12	Yes
Voice Command Support	Allow hands-free destination selection.	No	Medium	Not Started	10	Yes
Environmental Lighting Adaptation	Adjust AR visuals to lighting conditions.	Yes	Medium	In Progress	8	Yes
Navigation Accuracy Evaluation	Measure localization and path accuracy.	Yes	Medium	In Progress	10	Yes
Battery & Resource Optimization	Reduce power and CPU/GPU usage.	Yes	High	Not Started	8	Yes
Mid-Project DEMO Preparation	Prepare demo for stakeholders.	Yes	High	Complete	6	Yes
SPRINT 2	No	Yes	High	Complete	141	Yes

SLAM Algorithm Integration	Enable real-time localization and mapping.	Yes	High	In Progress	20	Yes
Camera & IMU Sensor Fusion	Combine camera and IMU for stability.	Yes	High	In Progress	14	Yes
AR Path Visualization	Overlay arrows and navigation cues.	Yes	High	In Progress	16	Yes
Indoor Map Data Structure	Define building data representation.	Yes	Medium	In Progress	10	Yes
Destination Search Module	Search locations.	Yes	High	In Progress	12	Yes
Path Planning Algorithm	Compute optimal route.	Yes	High	In Progress	15	Yes
Dynamic Route Update	Recalculate route.	Yes	High	In Progress	12	Yes
Multi-Floor Navigation	Support stairs and elevators.	Yes	Medium	In Progress	10	Yes
AR UI Design	Simple interface.	Yes	Medium	In Progress	8	Yes
Performance Optimization	Maintain 60 FPS.	Yes	High	In Progress	10	Yes
System Integration Testing	Test full system.	Yes	High	Not Started	8	Yes
Error Handling	Prevent crashes.	Yes	Medium	In Progress	6	Yes
SPRINT 1	No	Yes	High	Complete	24	Yes
Requirement Analysis and Use Case Definition	Gather functional and non-functional requirements and define scope.	Yes	High	Complete	4	Yes

Establish Contact with Potential Partner	Reach out for feedback/sponsorship.	Yes	High	Complete	12	Yes
Technology Stack Selection	Choose AR/SLAM tools and setup environment.	No	Medium	Complete	8	Yes
Preliminary SLAM Feasibility Test	Initial experiments using sensors.	No	Medium	Complete	0	Yes